Sum-Of-Squares Programming for Ma-Trudinger-Wang Regularity of Optimal Transport Maps

Abhishek Halder

Department of Aerospace Engineering, Iowa State University
Department of Applied Mathematics, University of California Santa Cruz

Joint work with Sachin Shivakumar (Los Alamos National Lab), Georgiy A. Bondar (UC Santa Cruz), Gabriel Khan (J.P. Morgan Chase)

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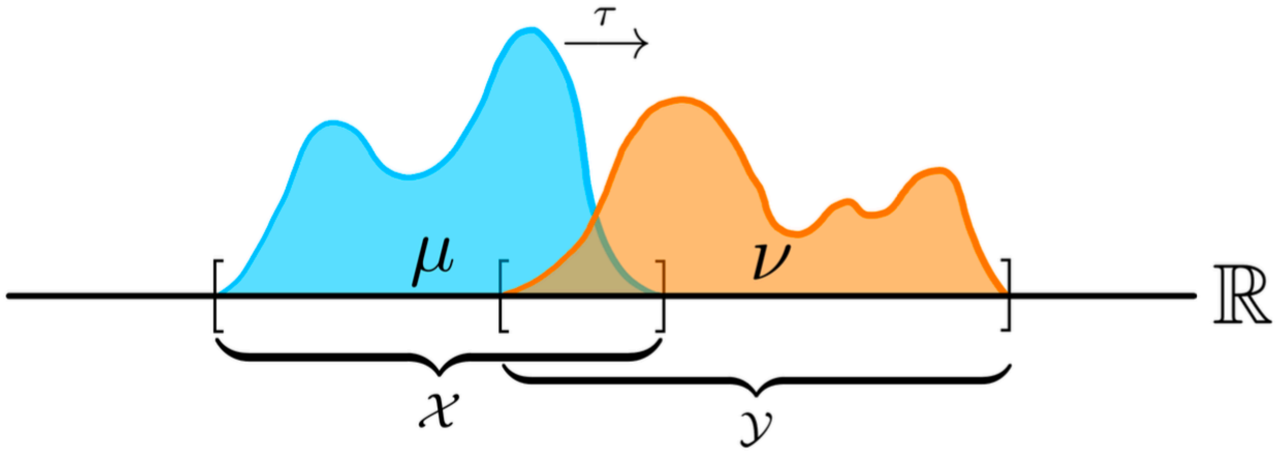
OT Map



$\mathcal{X}, \mathcal{Y} \subseteq \mathcal{M}$, smooth, compact, connected n dimensional manifold

Sufficiently smooth $c : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$

$$\tau_{\text{opt}} = \arg\inf_{\tau:\mathcal{X}\to\mathcal{Y}} \int_{\mathcal{X}} c(x, \tau(x)) d\mu(x)$$

$$\text{subject to} \quad \tau_{\#}\mu = \nu$$

OT map

Ground cost

Monge formulation, 1781

Nonlinear nonconvex program

Regularity of the OT Map: Motivation

Regularity (e.g., continuity, injectivity) implies

- stability: nearby points go nearby. Better interpretability of OT map
- structured variational/PDE tools can be applied
- possibility to design efficient numerical algorithms adapted to the geometry

ML/engineering applications of OT with non-Euclidean manifolds or costs

Manifold	$c(x,y)$	**Application**
$\mathbb{S}^{n-1} \times \mathbb{S}^{n-1}$	$-\log \lVert x-y \rVert$	Reflector Antenna
$\mathbb{S}^{n-1} \times \mathbb{S}^{n-1}$	$b_1 - \sqrt{b_2 + b_3 \lVert x-y \rVert^2}$	Reflector Antenna
$\mathbb{R}^n \times \mathbb{S}^{n-1}$	$-\langle x,y \rangle / \lVert x \rVert$	Text-to-image Generation
$\mathbb{H}^n \times \mathbb{H}^n$	$-\cosh \circ d_{\mathbb{H}^n}(x,y) = -\left(1 + 2\frac{\lVert x-y \rVert^2}{(1-\lVert x \rVert^2)(1-\lVert y \rVert^2)}\right)$	Word Embeddings
$\mathbb{R}^n \times \mathbb{R}^n$	$\inf_{\gamma \in \mathcal{C}(x,y)} \int_0^1 \mathcal{L}(\gamma_t, \dot{\gamma}_t)\, \mathrm{d}t$	Diffusion Models
Unknown	$\frac{1}{2} d^2(x,y)$ learnt from data	??
$\mathbb{R}^n \times \mathbb{R}^n$	$\log(1 + \sum_{i=1}^n \exp(x_i - y_i))$	Information Theory

Regularity of the OT Map: Background

Question: is τ_{opt} continuous?

Answer: Yes if μ, ν abs. continuous + extra condition on c and manifold

Regularity of the OT Map: Background

Question: is τ_{opt} continuous?

Answer: Yes if μ, ν abs. continuous + extra condition on c and manifold

Defn: Ma-Trudinger-Wang (MTW) tensor (2005, 2009)

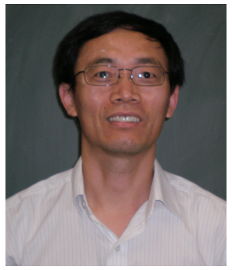
$$\mathfrak{S}_{(x,y)}(\xi, \eta) := \sum_{i,j,k,l,p,q,r,s} (c_{ij,p}c^{p,q}c_{q,rs} - c_{ij,rs})c^{r,k}c^{s,l}\xi_i\xi_j\eta_k\eta_l$$

$$\forall\, x \in \mathcal{X}, y \in \mathcal{Y}, \xi \in T_x\mathcal{X}, \eta \in T_y^*\mathcal{Y}$$

$$c_{ij,kl} = \partial_{x_i}\partial_{x_j}\partial_{y_k}\partial_{y_l}c(x,y), \quad c^{i,j}(x,y) = \left[((\nabla_x \otimes \nabla_y)c)^{-1}\right]_{i,j}$$

X-N. Ma

N. Trudinger



X-J. Wang

Regularity of the OT Map: Background

Defn: MTW(κ), $\kappa \geq 0$

If $\exists k \geq 0$ s.t. $\mathfrak{S}_{(\cdot,\cdot)}(\xi, \eta) \geq \kappa \|\xi\|^2 \|\eta\|^2 \; \forall (x, y) \in \mathcal{X} \times \mathcal{Y}$,

$\forall$orthogonal pairs (η, ξ), i.e., $\eta(\xi) = 0$

then c satisfies MTW(κ)

Defn: Nonnegative Cost Curvature (NNCC)

If $\mathfrak{S}_{(\cdot,\cdot)}(\xi, \eta) \geq 0 \; \forall (\xi, \eta)$ then c satisfies NNCC

Regularity of the OT Map: Background

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Defn: Nonnegative Cost Curvature (NNCC)

If $\mathfrak{S}_{(\cdot,\cdot)}(\xi, \eta) \geq 0 \ \forall (\xi, \eta)$ then c satisfies NNCC

Difficult to verify analytically, also difficult to generalize

"At present, there does not seem to be any convincing idea around to attack this problem". C. Villani, *J. Functional Analysis*, 255(9), p. 2686, 2008.

Proposed Approach: Computational Certificate

Main idea: Sum-of-squares (SOS) tightening of non-negativity conditions

Assumption A1: MTW tensor is rational in $(x,y) \in \mathcal{X} \times \mathcal{Y}$ semialgebraic

$$T_x\mathcal{X}, T_y^*\mathcal{Y} \cong \mathbb{R}^n \quad \Longrightarrow \quad \mathfrak{S}_{(x,y)}(\xi,\eta) = (\xi \otimes \eta)^\top \overbrace{F(x,y)}^{\in \mathbb{R}^{n^2 \times n^2}}(\xi \otimes \eta)$$

$$[F(x,y)]_{i+n(j-1),k+n(l-1)} = \sum_{p,q,r,s} (c_{ij,p}c^{p,q}c_{q,rs} - c_{ij,rs})c^{r,k}c^{s,l}$$

$$F = \frac{F_N}{F_D} \in \mathbb{R}_{N,D}[x,y], N, D \in \mathbb{N}$$

Sufficient but not necessary: c is rational in $(x,y) \in \mathcal{X} \times \mathcal{Y}$ semialgebraic

Problem Formulation

Assumption A1: MTW tensor is rational in $(x, y) \in \mathcal{X} \times \mathcal{Y}$ semialgebraic

Sufficient but not necessary: c is rational in $(x, y) \in \mathcal{X} \times \mathcal{Y}$ semialgebraic

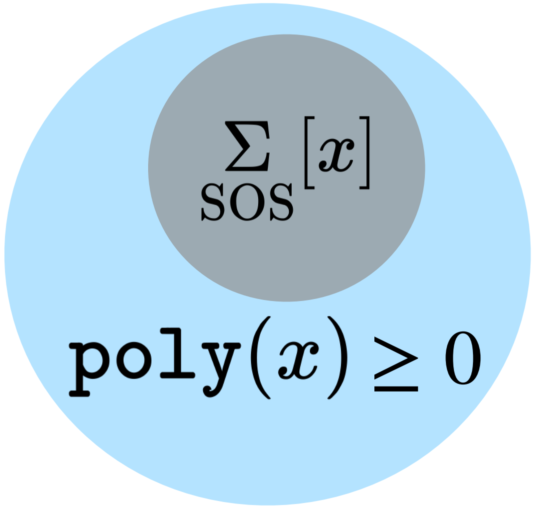
Forward problem: [Global certificate]

Given $c, \mathcal{X}, \mathcal{Y}$ as per **A1**, certify/falsify if the ground cost $c : \mathcal{X} \times \mathcal{Y} \mapsto \mathbb{R}_{\geq 0}$ satisfies either MTW(κ) or NNCC condition

Inverse problem: [Local certificate]

Given $c, \mathcal{X}, \mathcal{Y}$ as per **A1**, find semialgebraic $\mathcal{U} \times \mathcal{V} \subseteq \mathcal{X} \times \mathcal{Y}$ such that $c : \mathcal{U} \times \mathcal{V} \mapsto \mathbb{R}_{\geq 0}$ satisfies either MTW(κ) or NNCC condition

Recap: SOS Programming



SOS decomposition:

$$s \in \sum_{\text{SOS}}[x] \;\Rightarrow s = Z_d(x)^\top S Z_d(x) \text{ for some } S \succeq 0,\; d \in \mathbb{N}$$

where $Z_d(x)$ with monomial entries $(1, x, \ldots, x^d)$ of length $\zeta := \sum_{r=0}^{d} \binom{n+r-1}{r}$

Polynomial optimization [computationally intractable]: For $f, g_{i\in[n_g]} \in \mathbb{R}[x]$

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{such that} \quad x \in \mathcal{C} := \{x \in \mathbb{R}^n \mid g_i(x) \leq 0 \forall i \in [\![n_g]\!]\}$$

$$\Updownarrow$$

$$\max_{\gamma\in\mathbb{R}} \gamma \quad \text{subject to} \quad f(x) - \gamma \geq 0 \quad \forall x \in \mathcal{C} \text{ semialgebraic}$$

Recap: SOS Programming

$$f(x) - \gamma \geq 0 \quad \forall x \in \mathcal{C} \text{ Archimedean}$$

$$\Updownarrow \quad \text{Putinar's Positivstellensatz (1993)}$$

$$\exists s_0, s_1, \ldots, s_{n_g} \in \sum_{\text{SOS}}[x] \text{ such that } f(x) - \gamma = s_0(x) - \sum_{i \in [n_g]} s_i(x) g_i(x)$$

SOS tightening of polynomial optimization $\rightsquigarrow$ Semidefinite program (SDP)

$$\max_{(\gamma, S_0, S_1, \ldots, S_{n_g}) \in \mathbb{R} \times \underbrace{\mathbb{S}_+^{\zeta} \times \ldots \times \mathbb{S}_+^{\zeta}}_{n_g+1 \text{ times}}} \gamma$$

$$\text{subject to} \quad f(x) - \gamma = Z_d(x)^\top S_0 Z_d(x) - \sum_{i \in [n_g]} Z_d(x)^\top S_i Z_d(x) g_i(x) \geq 0$$

Off-the-shelf SDP software: SOSTOOLS, YALMIP, SOSOPT

SOS Reformulation of the Forward Problem

$$\mathcal{X} \times \mathcal{Y} = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^n \mid m_i(x, y) \leq 0,\ m_i(x, y) \in \mathbb{R}_{d_m}[x, y]\ \forall i \in [\![\ell]\!]\}$$

NNCC certificate

If there exist SOS multipliers s_i such that

$$\left(F_N(x, y) + F_N^\top(x, y)\right) - s_0(x, y) F_D(x, y) + \sum_{i \in [\![\ell]\!]} s_i(x, y) m_i(x, y) \in \sum_{\text{sos}}^{n^2} [x, y]\ ,$$

then c satisfies NNCC on $\mathcal{X} \times \mathcal{Y}$.

MTW(κ) certificate

If there exist SOS multipliers s_i and an equality multiplier t such that

$$\begin{aligned}(\xi \otimes \eta)^\top F_N(x, y)(\xi \otimes \eta) - \kappa F_D(x, y) \left\|\xi\right\|^2 \left\|\eta\right\|^2 \\ + \sum_i s_i(x, y, \xi, \eta) m_i(x, y) + t(x, y, \xi, \eta) \eta^\top \xi \in \Sigma[x, y, \xi, \eta],\end{aligned}$$

then c satisfies $\mathrm{MTW}(\kappa)$.

SOS Reformulation of the Inverse Problem

Parameterize the certified region

$$\mathcal{U} \times \mathcal{V} = \{(x, y) \in \Lambda : V(x, y) \leq 0\},$$

where Λ is compact and V is a polynomial function (decision variable).

Original Problem

$$\begin{aligned} \underset{\mathcal{U}\times\mathcal{V}\subseteq\mathcal{X}\times\mathcal{Y}}{\arg\max} \quad & \operatorname{vol}(\mathcal{U} \times \mathcal{V}) \\ \text{subject to} \quad & \mathfrak{S}_{(u,v)}(\xi, \eta) \geq 0, \quad \forall (u, v) \in \mathcal{U} \times \mathcal{V},\ \xi \in T_u\mathcal{U},\ \eta \in T_v^*\mathcal{V}. \end{aligned}$$

Sublevel-set Reformulation

$$\begin{aligned} \max_{V\in\mathbb{R}_d[x,y]} \quad & \operatorname{vol}\{(x, y) \mid V(x, y) \leq 0\} \\ \text{subject to} \quad & m_i(x, y) \leq V(x, y) \quad \forall\, i \in \ell. \\ & V(x, y) + \mathfrak{S}_{(x,y)}(\xi, \eta) \geq 0 \geq 0, \quad \forall (x, y) \in \mathcal{X} \times \mathcal{Y},\ \xi \in T_x\mathcal{X},\ \eta \in T_y^*\mathcal{Y}. \end{aligned}$$

Volume Proxy for the Inverse Problem

Parameterize the certified region

$$\mathcal{U} \times \mathcal{V} = \{(x, y) \in \Lambda : V(x, y) \leq 0\},$$

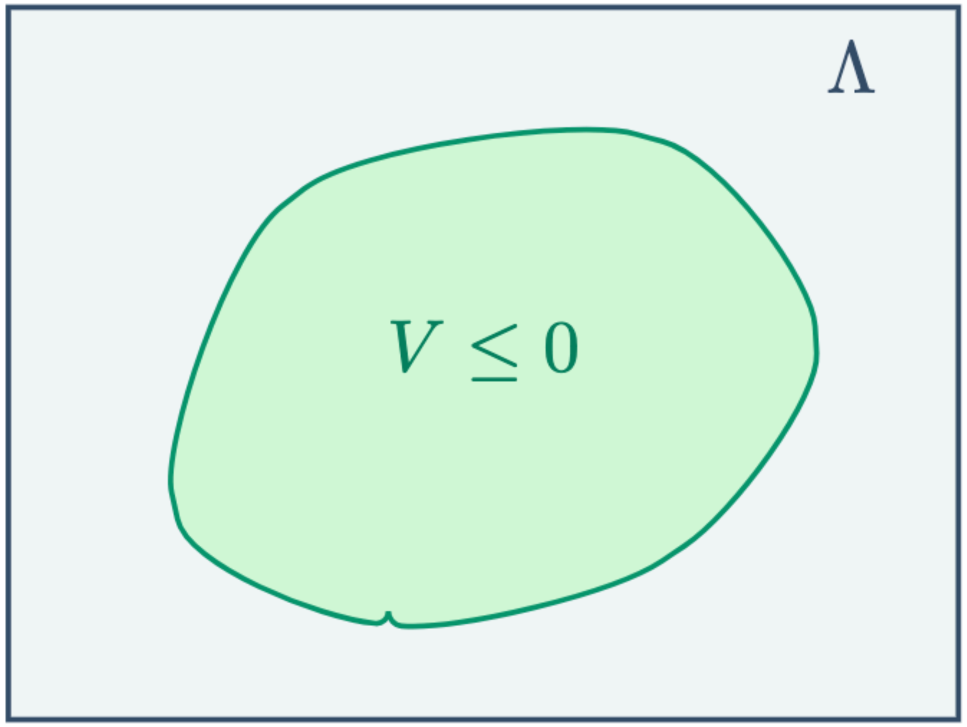
where Λ is a compact search box and V is a polynomial decision variable.

Volume surrogate

Instead of directly maximizing volume, minimize

$$\int_{\Lambda} V(x, y)\, dx dy$$

subject to SOS constraints that force the zero sublevel set to lie inside the regularity region.



Numerical Example 1: Forward Problem

Perturbed Euclidean ground cost

$$c(x,y) = \|x-y\|_2^2 - \varepsilon\|x-y\|_2^4, \quad x,y \in \mathbb{R}^n, \varepsilon > 0$$

Lee & Li (2009): for ε small enough, MTW(0) holds on

$$\mathcal{X} \times \mathcal{Y} := \{(x,y) \in \mathbb{R}^n \times \mathbb{R}^n \mid \|x-y\|_2 \leq 0.5\}$$

But how small is small enough?

We used SOS SDP + bisection to find $\varepsilon_{\max}$ such that MTW(0) holds

Dimensions, n	1	2
$\varepsilon_{\max}$	$0.67(\approx \frac{2}{3})$	$1.05 \cdot 10^{-2}$
Residual	$1.19 \cdot 10^{-7}$	$4.18 \cdot 10^{-7}$

in Lee & Li (2009) via nontrivial analysis

No analytical computation known

Numerical Example 2: Forward Problem

Log partition ground cost (Pal & Wong, 2018; Khan & Zhang, 2020) from stochastic portfolio theory

$$c(x,y) = \Psi_{\mathrm{IsoMulNor}}(x-y),\ \Psi_{\mathrm{IsoMulNor}}(x) := \overbrace{\frac{1}{2}\left(-\log x_1 + \sum_{i=2}^{n} x_i^2/x_1\right)}^{\textbf{not rational!}}$$

Our method still applies because

$$\mathfrak{S}_{(x,y)}(\xi,\eta) \propto \mathfrak{A}_x(\xi,\eta) = \mathrm{poly}(x,\xi,\eta)/x_1^2\xi_n^2$$

No analytical method exists to verify MTW(0) for $n > 2$

We can verify MTW(0) for $\mathcal{X} = \mathcal{Y} = \{x \in \mathbb{R}^n \mid x_1 > 0\}\ \forall n \in \mathbb{N}$

Dimensions, n	3	4	5	6
Residual	$1.034 \cdot 10^{-7}$	$4.804 \cdot 10^{-8}$	$4.683 \cdot 10^{-8}$	$3.475 \cdot 10^{-11}$
Total time (s)	0.7220	0.8050	1.2520	1.6690

Numerical Example 2: Forward Problem (contd.)

For $n = 3$, our method discovered SOS decomposition:

$$\text{poly}(x, \xi, \eta) = s(x, \xi, \eta)^\top s(x, \xi, \eta)$$

$$s(x,\xi,\eta) = \begin{bmatrix} 0 & -1.4 & 0 & 0.24 & 0 & 0 \\ 2.4 & 0 & -0.17 & 0 & 0 & 0 \\ 0 & 1.4 & 0 & -0.24 & 0 & 0 \\ -2.4 & 0 & 0.17 & 0 & -0.0002 & 0 \\ 0 & -1.4 & 0 & 0.25 & 0 & -1.2 \\ 2.4 & 0 & -0.17 & 0 & -0.0002 & 0 \\ -1.6 & 0 & -1.9 & 0 & 0 & 0 \\ 0 & 0.52 & 0 & 1.3 & 0 & 0 \\ 0 & 1.4 & 0 & -0.25 & 0 & 0 \\ -0.84 & 0 & 2 & 0 & 0 & 0 \\ 0 & -0.52 & 0 & -1.3 & 0 & 0 \end{bmatrix} \begin{bmatrix} \eta_1\xi_1^2\xi_2 \\ \eta_1\xi_1^2\xi_3 \\ \eta_1\xi_1\xi_2\xi_1 \\ \eta_1\xi_1\xi_3\xi_1 \\ \eta_2\xi_1\xi_2\xi_2 \\ \eta_2\xi_1\xi_2\xi_3 \end{bmatrix}$$

Numerical Example 3: Inverse Problem

Perturbed Euclidean ground cost

$$c(x,y) = \|x-y\|_2^2 - \varepsilon\|x-y\|_2^4, \quad x,y \in \mathbb{R}^n, \varepsilon > 0$$

Lee & Li (2009): for $\varepsilon \gg 0$, MTW(0) fails on

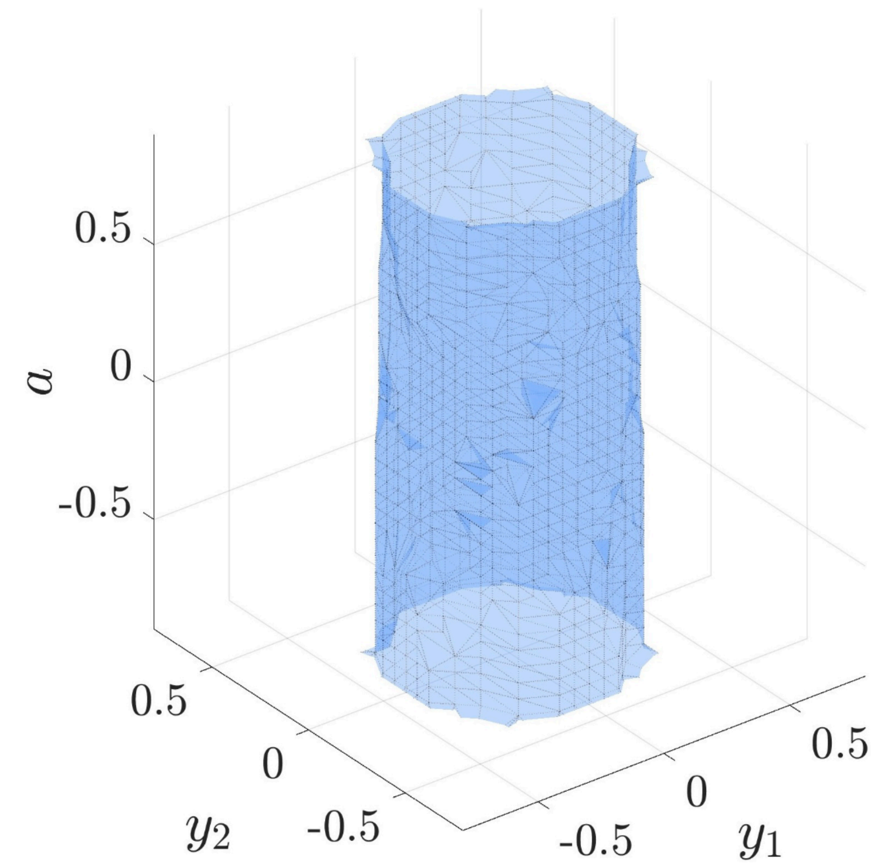
$$\mathcal{X} \times \mathcal{Y} := \{(x,y) \in \mathbb{R}^n \times \mathbb{R}^n \mid \|x-y\|_2 \leq 0.5\}$$

Fix $\varepsilon = 1, \Lambda = [-1,1]^2, \mathcal{X} = \{[0,0]\}$

Parameterize $(\xi,\eta) = \left([a,1]^\top, [-1,a]^\top\right)$

What the inverse SOS returns

A degree-14 polynomial sublevel set $V \leq 0$ certifying a region where the MTW tensor is nonnegative.



Inner approx. of region where MTW(0) tensor ≥ 0, solve time 0.97 s

Numerical Example 4: Inverse Problem

Non-Euclidean geometry: Squared distance cost for a surface of positive Gaussian curvature

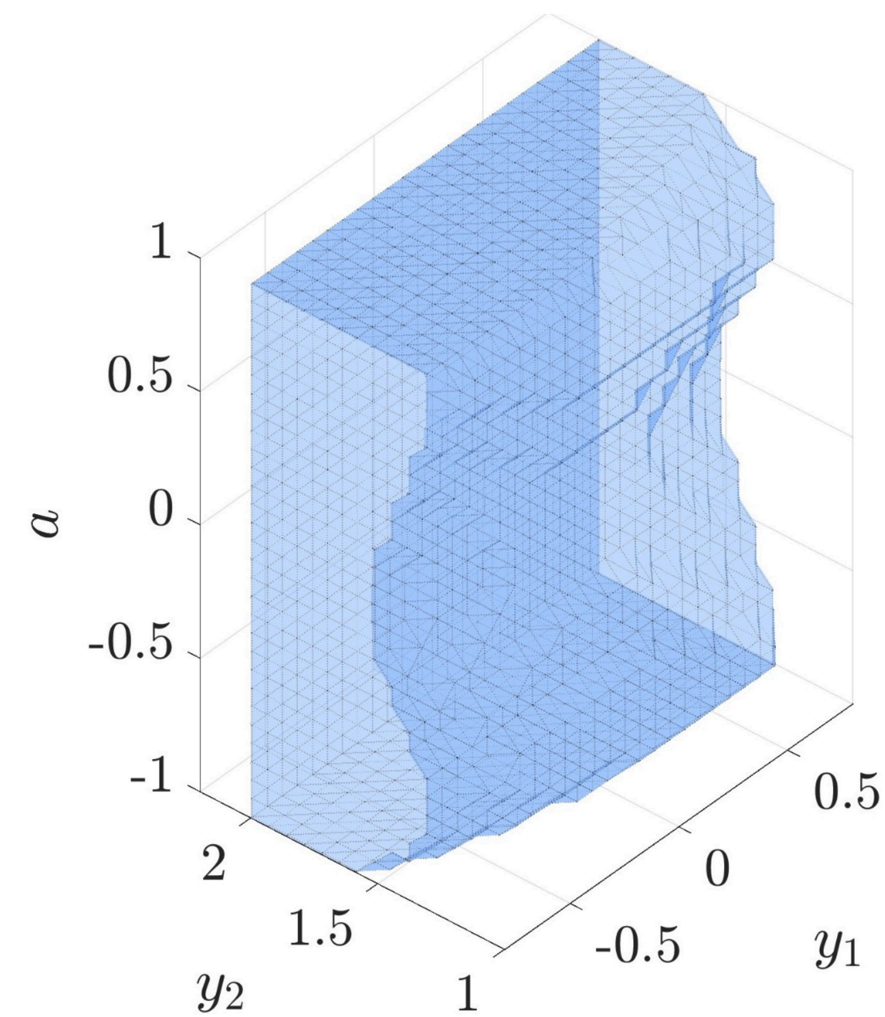
$$c(x,y) = 3(x_1 - y_1)^2(x_2 + y_2) + 4(x_2^3 + y_2^3) - (4x_2y_2 - (x_1 - y_1)^2)^{\frac{3}{2}}$$

MTW(0) holds around $\{x = y\}$

Parameterize $(\xi, \eta) = \left([a, 1]^\top, [-1, a]^\top\right)$

Symmetries reduce the computational load

Change-of-variable removes square root



Inner approx. of region where MTW(0) tensor ≥ 0, solve time 19.6 s

Computational Complexities

Both NNCC and MTW: sub-quadratic in # of constraints, polynomial in cost degree, exponential in manifold dimension

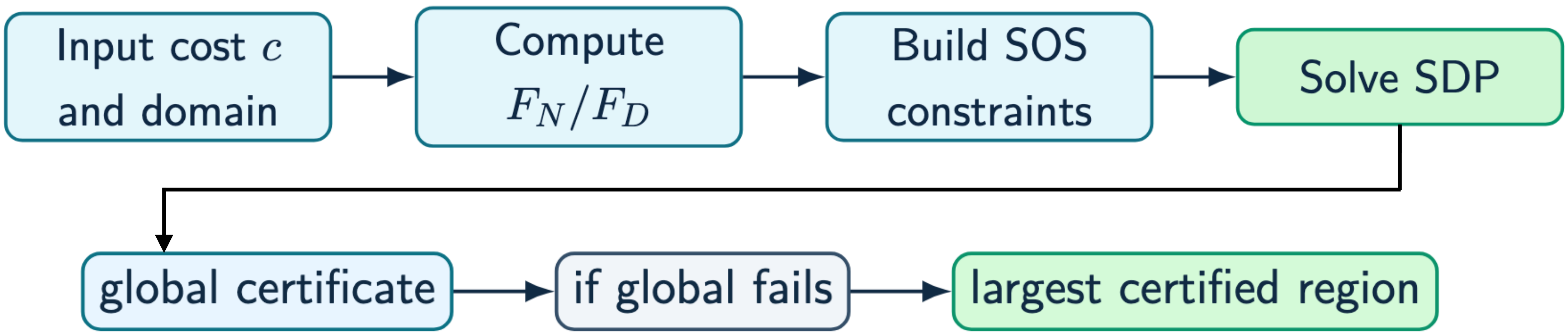
MTW more computation heavy than NNCC

Detailed analysis in JOTA paper

Leveraging symmetries reduce computation: require expertise and insight

Outlook

Summary of proposed method:



Current limitations/opportunities for future work:

- sufficient: SOS failure is not informative

- we did not optimize the SOS code to exploit problem-specific sparsity

- inner approximation from volume proxy is not theoretically maximal

Thank You